RI2ACT Research Idea 2 Action · PLNYRWRKWRKYRPLN

Small Group Exercise

Objective: Draft a research question that addresses a gap in pediatric firearm injury prevention using currently available data sources and your own interests.

Name / project: Date:

Part 1: Define Interests

Populations of Interest

- ☐ Children under 12
- ☐ Adolescents (12–17)
- ☐ School age youth (5–17)
- ☐ Racial/ethnic minority groups
- ☐ Urban youth
- ☐ Rural youth
- ☐ LGBTQ+ youth
- ☐ Fatal injury
- ☐ Nonfatal injury
- ☐ Unintentional injury
- ☐ Assault/Homicide
- ☐ Self-Harm/Suicide
- ☐ School-based violence
- ☐ Domestic violence
- ☐ Location (national, etc.)
- ☐ Other

Variables of Interest: interventions, factors, outcome

- ☐ Mental health
- ☐ Hospital utilization
- ☐ Violence Intervention
- ☐ Recurrence or reinjury
- ☐ Community factors
- ☐ School safety
- ☐ Policy evaluation/Laws
- ☐ Access to firearms/safe storage
- ☐ Lethal Means Counseling
- ☐ Provider/Trainee Education
- ☐ Clinical care
- ☐ Community Based Intervention
- ☐ Hospital Based Interventions
- ☐ Advocacy
- ☐ Awareness
- ☐ Qualitative data
- ☐ Other

Part 2: Identify a Gap

- ☐ Longitudinal national data on nonfatal firearm injuries in children
- ☐ Long-term outcomes after injury. Any specific outcomes?
- ☐ Gaps in knowledge (provider or patient) about household firearm access and storage
- ☐ Disparities by race, income, neighborhood, etc.
- ☐ Evaluations of prevention programs or policies: Which one?
- ☐ Risk factors for firearm injury. Any specific risk factors?
- ☐ How do changes in perceived community connectedness and collective efficacy influence trends in firearm violence over time?
- ☐ Evaluate the impact of community-level protective factors (social trust, neighborhood engagement, local mutual aid networks) on firearm violence.

Authors: SJ Patel, A Silver, MK Goyal, E Fleegler, N Romo, A Hogan, G Azzarone, K Gastineau

☐ Predictors of poor outcomes after firearm injury (example: rates of mental health evaluations post injury). Which ones?

☐ Other

Part 3: Draft Your Research Question

Example: Among [Population], how does [Exposure or Policy] compared to [Comparison] affect [Outcome] over [Timeframe or Setting]?

Fill in the pieces (optional):

Population

Exposure or Policy

Comparison

Outcome

Timeframe or Setting

Research Question

Part 4: Identify Data Sources

See references for more information. Check the sources you might use and add your own notes.

	Data source	Key data	Limitations	Examples/Notes
☐	**Claims data (PHIS, HCUP, NEDS, see toolkit)**	Utilization with clinical details, PHIS with more contextual data	Costly, older data, PHIS only has Children's Hospitals	
☐	**CDC WISQARS**	Fatal and nonfatal injury data with mapping capabilities (available only for fatal data)	Updates delayed by 2–3 years. Use caution when describing nonfatal data by intent.	
☐	**CDC WONDER**	Fatal injury		
☐	**CDC NVDRS**	Highly detailed fatality data	Need to apply for restricted access	
☐	**NEISS-FISS**	ED firearm injuries, national sample	Narratives require advanced methods (AI, NLP) and special agreement with CDC	
☐	**RAND Firearm Hospitalizations**		Updated through 2025	
☐	**RAND state firearm ownership data**	State policy comparisons	Only up to 2016	
☐	**Gun Violence Archive**	Real-time incident-level data	Better for firearm fatalities vs injuries/suicide (reporting bias)	
☐	**EveryShot**		New, only has a few cities	

	Data source	Key data	Limitations	Examples/Notes
☐	**FASTER-AVERT**		New, limited at this time	
☐	**ACS, SVI, COI, etc.**	Neighborhood socioeconomic data		
☐	**RAND Firearm Law Database**		Only up to 2016	
☐	**Statefirearmlaws.com (Siegel's Database)**	Highly detailed state firearm laws from 1976–2019 + estimates of state level firearm ownership per year	Off line for 3 weeks	
☐	**Giffords**	State firearm laws – very up to date		
☐	**YRBSS**	Firearm carrying and risk behaviors in youth		
☐	**Trauma Registries**			
☐	**Coroner's Reports**			
☐	**Law Enforcement Data**	Police data	Incomplete outcome data and variables available	
☐	**Qualitative data**	Community, law enforcement, provider or parent/patient voice		*Mixed methods can help contextualize other epidemiologic findings.*
☐	**Other:**			

Part 5: Feasibility

1. Can you access the data source(s) needed?
   ☐ Yes ☐ Not yet – refine question, add to to-dos ☐ Unsure

2. Is your outcome measurable using this data?
   ☐ Yes ☐ No – refine question ☐ Unsure

3. Do you have the expertise needed for the data management and/or analysis?
   ☐ Yes ☐ No ☐ Unsure

4. Would funding be needed?
   ☐ Yes ☐ No ☐ Unsure

5. Would IRB approval be needed (e.g., for individual-level data)?
   ☐ Yes ☐ No ☐ Unsure

Concerns regarding current funding/political environment (data availability, funding, institutional priorities, etc.)

Part 6: Share & Refine

Share your draft question with a partner or your small group. Discuss feasibility and use feedback to refine it.

Final Version of Your Research Question

Part 7: Next Steps/Timeline for yourself

To do	What/Who do I need	Rough Deadline

Common Pitfalls to Avoid in Pediatric Firearm Injury Research

Check each one once you've considered it for your project.

- ☐ **Diversify Outcomes Beyond Mortality**
  Include nonfatal firearm injuries and exposures using claims data, trauma registries, or incident-level sources.
- ☐ **Disaggregate by Age**
  Avoid lumping all youth under age 20; consider stratification by developmental stages (e.g., 0–5, 6–12, 13–17). Consider pros/cons to 20–24 – purview of pediatrics, highest number of injuries/fatalities.
- ☐ **Incorporate Contextual Data**
  Use NVDRS, EveryShot, NEISS-FISS (if you can access), your EMR, or qualitative sources to understand the setting, intent, and circumstances of incidents.
- ☐ **Be Cautious with Policy Evaluations**
  Avoid ecological fallacy. Use local-level policy data and advanced methods like difference-in-differences. Recognize that many of these studies are ecological in nature (combining disparate datasets) but that doesn't make them incorrect or poor studies.
- ☐ **Address Structural Inequities**
  Include place-based and structural variables (Macy/Kendi Equity Framework). Don't just control for race, investigate upstream determinants.
- ☐ **Expand Beyond Immediate Injury Outcomes**
  Consider long-term effects like mental health, education, and community displacement.
- ☐ **Validate and Cross-Check Claims Data**
  Use multiple data sources to confirm firearm-related ICD-10 codes and ensure accuracy.